# JONATHAN MA

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CS student with 5+ years of experience in spacecraft software, embedded systems, and robotics. Skilled in C++, microcontrollers, and Python, with hands-on experience writing flight code for active space missions. Proven leader who thrives in fast-paced, interdisciplinary teams and is eager to tackle mission-critical problems.

## Education

**Cornell University, College of Engineering (Ithaca, NY)**

Master of Engineering, Computer Science *Anticipated December 2026*

Bachelor of Science, Computer Science (GPA 3.99, Dean's List) *Anticipated May 2026*

- Selected Coursework: Computer Vision, Machine Learning, Systems/OS Programming, Robotics and RL
- TA for Computer Organization and Systems (C, Assembly), OCaml Programming, and Intro to Robotics classes

## Technical Skills

- Languages (Proficient): Python, Java, C++, C, JavaScript, TypeScript, HTML/CSS, OCaml, PostgreSQL
- Frameworks: Arduino, ROS, OpenCV, NumPy, Pandas, Flask, React, JavaFX, Flutter, Angular, Firebase
- Tools: Git, Docker, Linux, Bash, PlatformIO, Ansible, Jupyter, Elasticsearch, Kibana, Jira, CI/CD, Heroku

## Experience

**SpaceX: Flight Software Engineering Intern (Starshield)** *Summer 2025*

- Architected multi-system flight and ground software solution to deconflict satellite ops in embedded Linux env
- Led design review w/ 10+ stakeholders, converting ambiguous guidelines to actionable tasks for robust solution
- Overhauled satellite integration and testing Python script to increase modularity and testability, taking ownership to drastically reduce risk for critical path test campaign while saving 1000s of engineering hours

**Cornell Space Systems Design Studio: Alpha CubeSat Flight Software Lead** *Oct 2022 - Present*

- Promoted to lead of 5 person team after 2 yrs for exemplary performance and initiative to learn/solve problems
- Used Arduino/C++ to build real-time embedded flight software for 1U CubeSat & light sail payload in LEO
- Developed code for memory constrained microcontrollers to interface with GPS, IMU, and LoRa radio avionics
- Conducted rigorous and rapid hardware-in-the-loop integration tests to ensure mission readiness
- Built & deployed full-stack ground station with intuitive React UI + Elasticsearch/Kibana telemetry dashboards

**RTX Collins Aerospace: Systems Engineering Intern (SEPP)** *Summer 2024*

- Collaborated with team of interns to conduct computer networking research studies for Info Centric Routing over IP library with Raspberry Pi fleet and extended 50K line open-source C++ codebase to support persistency

**Johns Hopkins University Applied Physics Lab: Ground Software Engineering Intern** *Summer 2023*

- Led team of interns through design and creation of Angular and Java based app for parsing binary spacecraft command and telemetry packets for NASA's IMAP and Dragonfly missions, used by 50+ employees
- Applied Agile methodologies and solicited feedback from project leads throughout project duration

**First Tech Challenge Robotics Team #7303: Robot Automation Lead** *Aug 2019 - June 2022*

- Won Maryland Tech Invite out of top 32 teams globally, Control Award for most innovative control/automation
- Collaborated with team to implement OpenCV obj detect, odometry localization, state machines & PID control
- Developed multistage automation and CV algorithms which can autonomously generate paths to collect and shoot rings while moving at 5 in target from anywhere on 12x12 ft field
- Created JavaFX simulator for path planning and replaying robot actions to enable testing w/o robot hardware

## Programming Projects

- Rubik's Cube Solving Robot (C, Python), Arduino powered robot optimized to solve Rubik's Cube in 3-4 sec
  - Created OpenCV pipeline to scan cube and create Rubik's cube mosaic, breadboarded Arduino circuit
  - Optimized stepper motor controller and improved solving algorithm efficiency by ~25x
- MirrorBot (C, Python), Developed mirror control algorithm which uses ROS, Arduino & Intel RealSense depth camera to detect faces and dynamically adjust reflections to encourage non-verbal interactions in public spaces
- Happiness App, Full-stack Flask, PostgreSQL, and React mood-tracking social media app with 100+ users